

CH1101: Chemistry I / Quantum Mechanics

Tutorial Sheet 01

Q1

Wavelength λ [m]	10^{-10}	10^{-8}	10^{-6}	10^{-4}	10^{-2}	1
Wavenumber $\tilde{\nu}$ [cm^{-1}]	10^8	10^6	10^4	100	1	0.01
Frequency ν [Hz]	3×10^{18}	3×10^{16}	3×10^{14}	3×10^{12}	3×10^{10}	3×10^8
Energy E [J mol^{-1}]	12×10^8	12×10^6	12×10^4	1200	12	12×10^{-2}

X-rays

UV-
visible

Infrared

Micro
waves

Radiowaves

$$E = h\nu = h \frac{c}{\lambda} = hc\tilde{\nu}$$

Let's do for one case,
X-rays

$$\lambda = 10^{-10} \text{ m} = 10^{-8} \text{ cm}$$

$$\tilde{\nu} = \frac{1}{\lambda} = \frac{1}{10^{-8} \text{ cm}} = 10^8 \text{ cm}^{-1}$$

$$h\nu = h \frac{c}{\lambda} = hc\tilde{\nu} \Rightarrow \boxed{\begin{matrix} \nu = c\tilde{\nu} \\ \nu \propto \tilde{\nu} \end{matrix}}$$

$$\begin{aligned}
 \nu &= c\tilde{\nu} \\
 &= (3 \times 10^{10} \text{ cm } \cancel{\text{s}^{-1}}) \times (10^8 \text{ cm}^{-1} \cancel{\text{s}}) \\
 &= 3 \times 10^{18} \text{ s}^{-1} = 3 \times 10^{18} \text{ Hz}
 \end{aligned}$$

$$\begin{aligned}
 E &= hc\tilde{\nu} \\
 &= (6.6 \times 10^{-34} \text{ J } \cancel{\text{s}}) \times (3 \times 10^{10} \text{ cm } \cancel{\text{s}^{-1}}) \times (10^8 \text{ cm}^{-1} \cancel{\text{s}}) \\
 &= 2.0 \times 10^{-15} \text{ J} \\
 &= (2.0 \times 10^{-15} \text{ J}) \times N_A \\
 &= (2.0 \times 10^{-15} \text{ J}) \times (6.0 \times 10^{23}) \text{ mol}^{-1} \\
 &= 12.0 \times 10^8 \text{ J mol}^{-1}
 \end{aligned}$$

Wavelength $\uparrow$
 Energy $\downarrow$

Frequency $\uparrow$
 Energy $\uparrow$